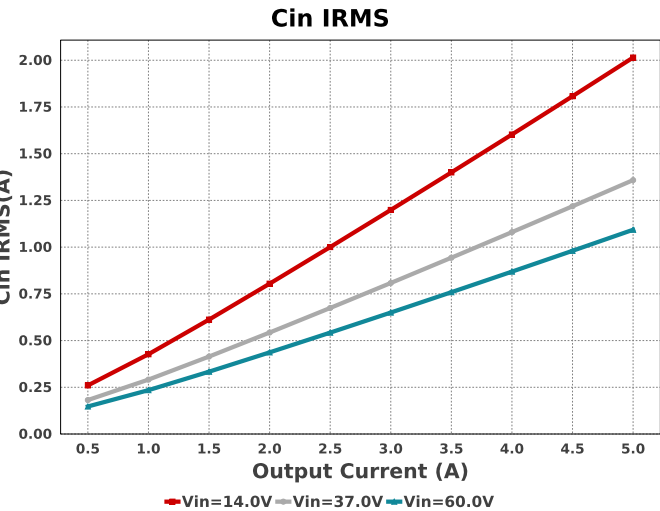
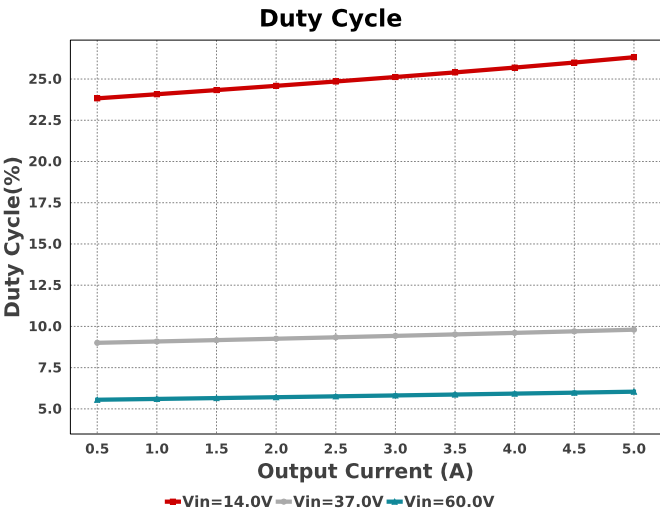
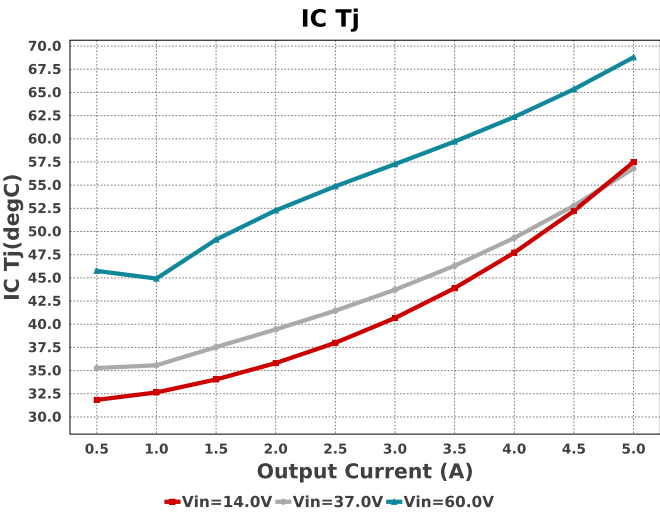
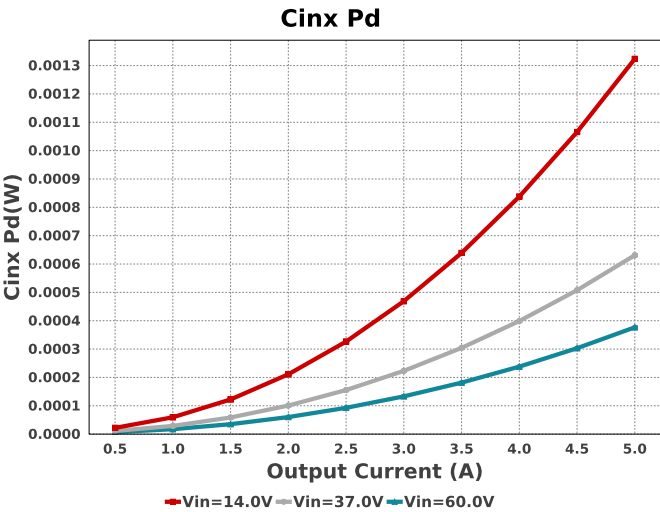
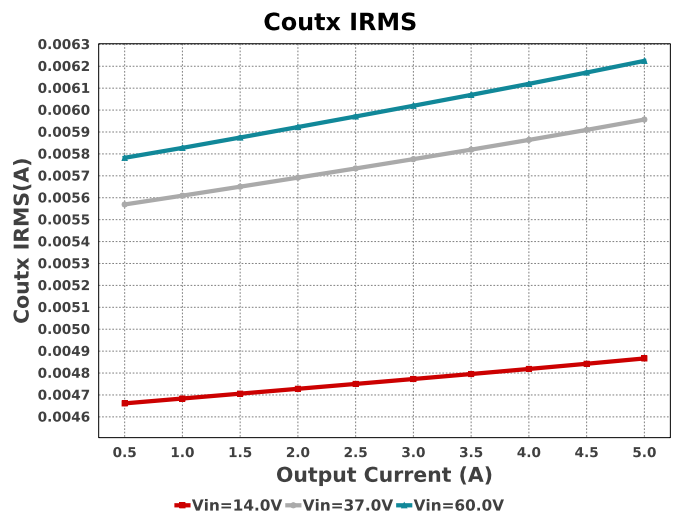
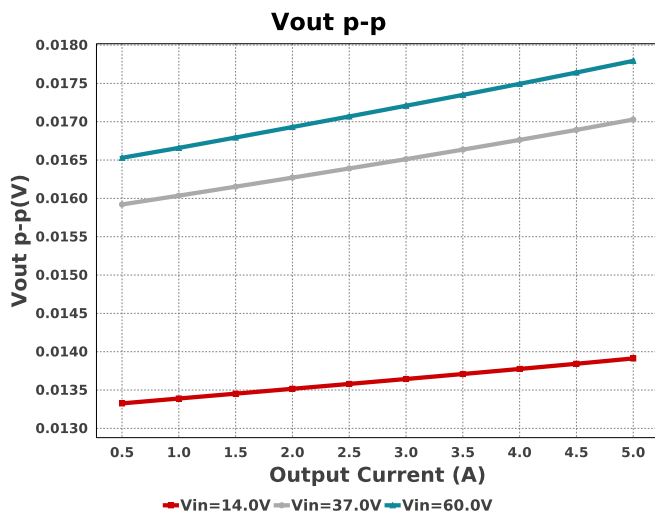
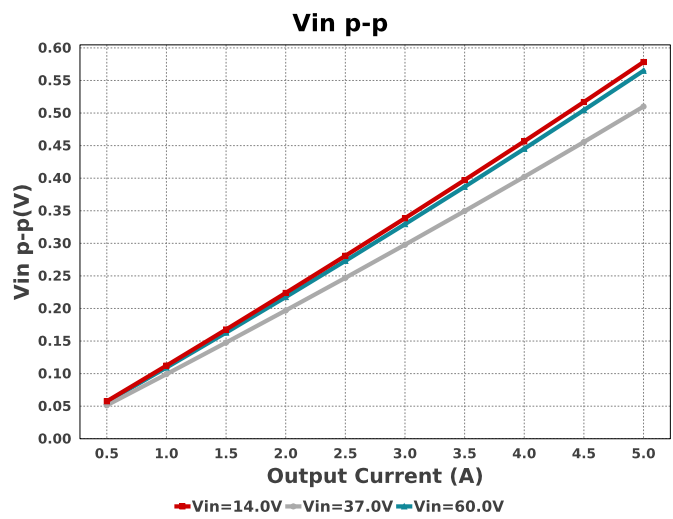
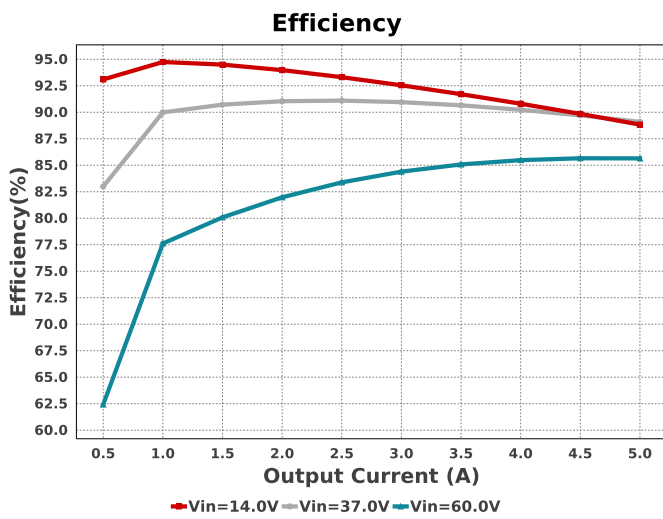
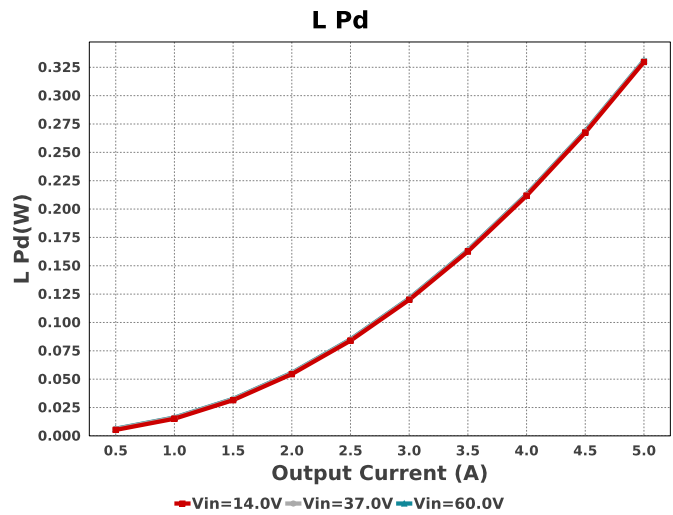
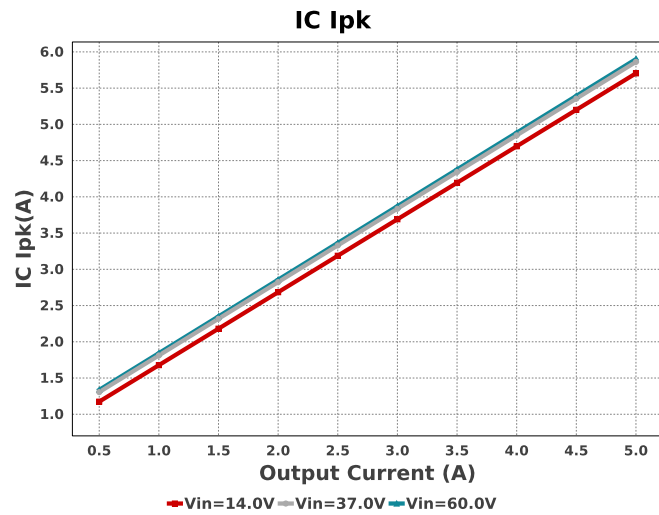


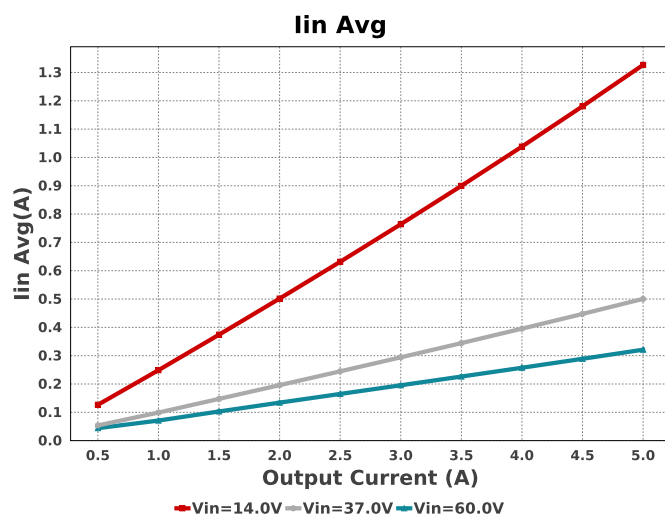
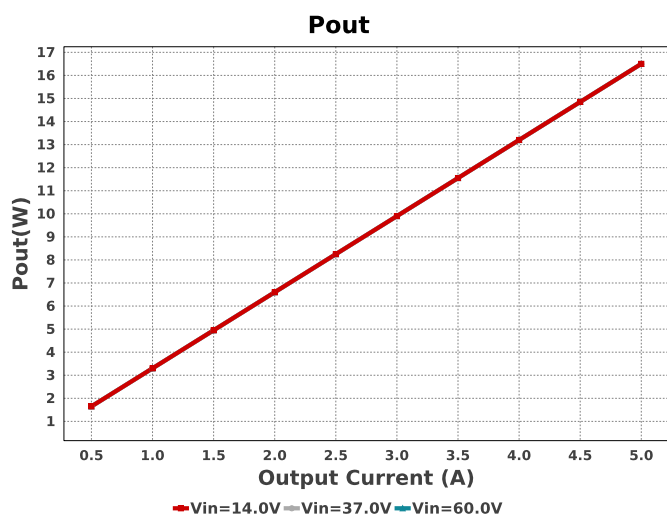
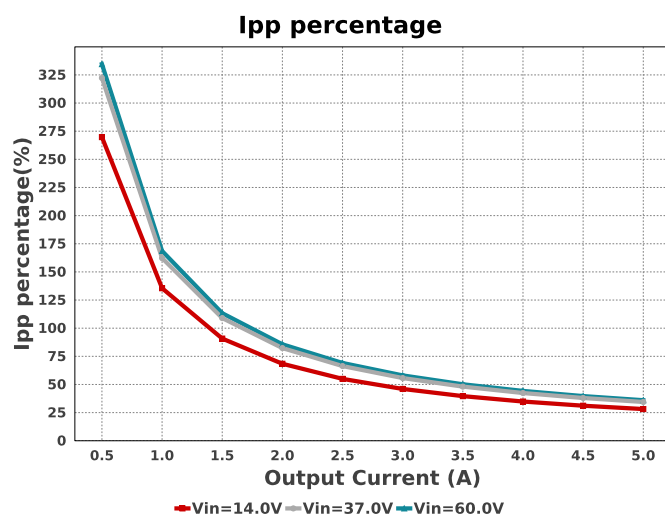
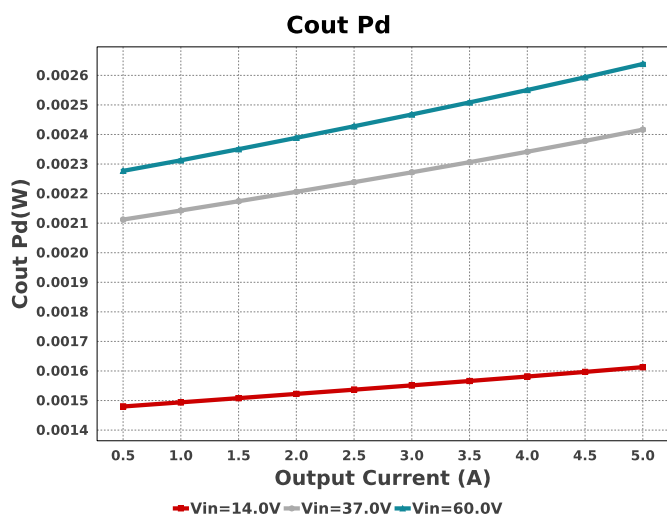
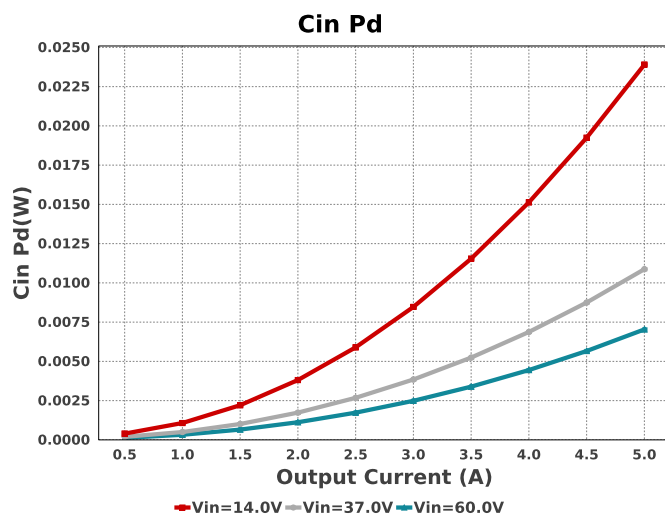
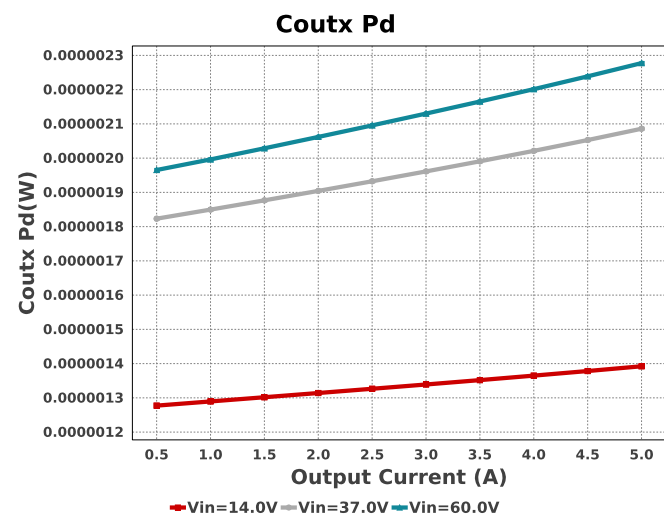


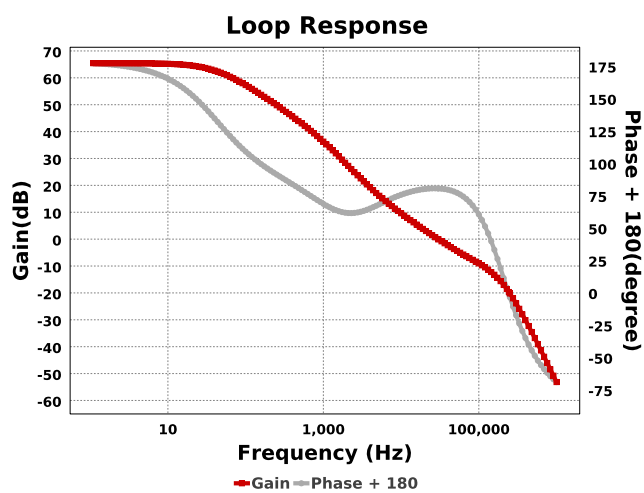
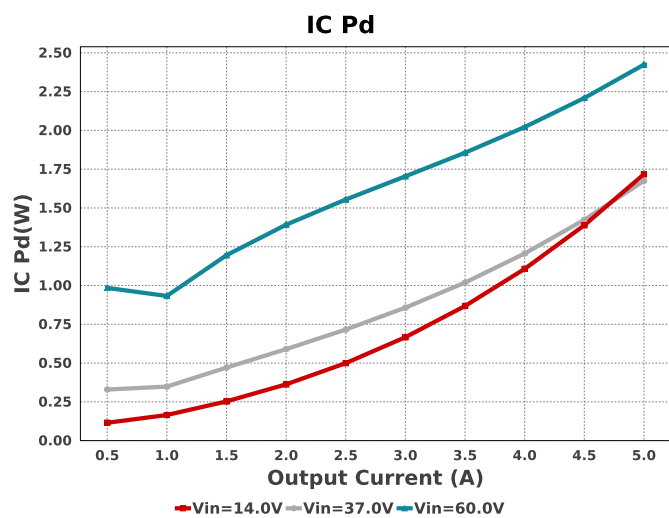
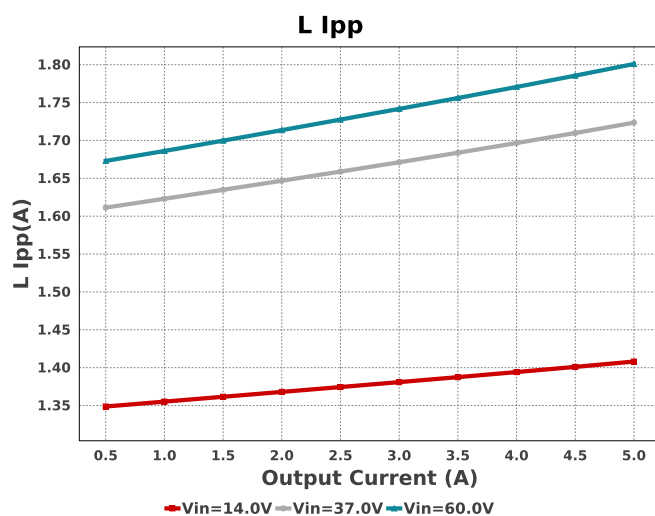
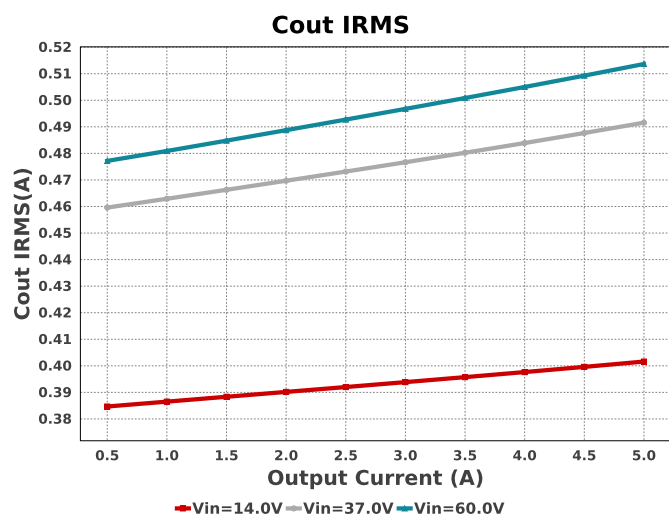
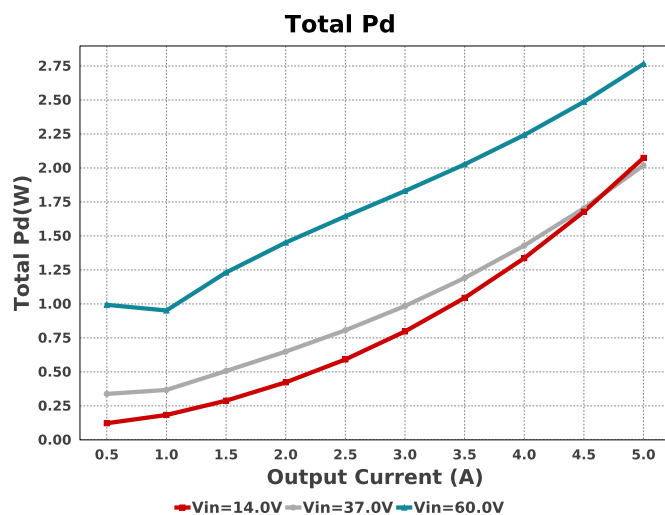
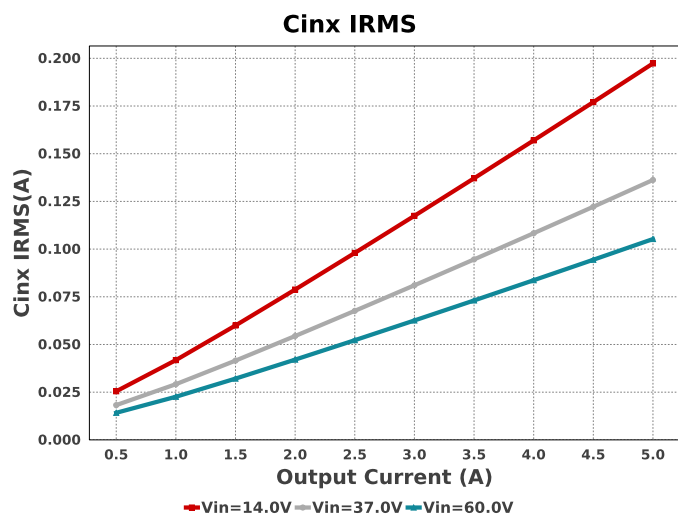
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Topology = Buck  
Created = 2025-10-31 23:52:18.255  
BOM Cost = \$4.23  
BOM Count = 13  
Total Pd = 2.77W

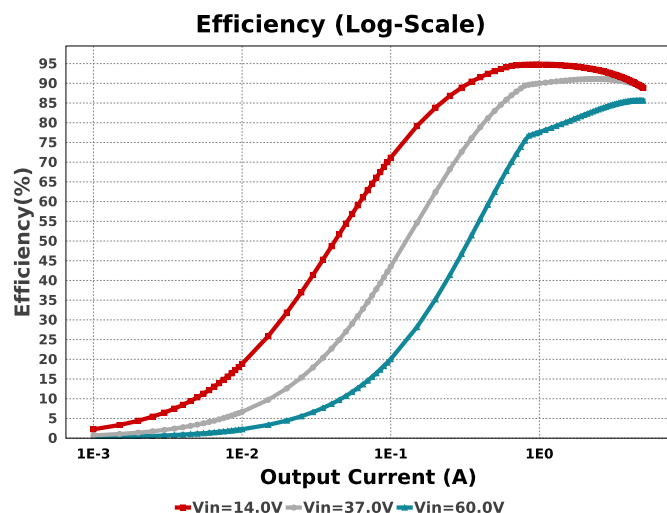
| Name   | Manufacturer      | Part Number                          | Properties                                            | Qty | Price  | Footprint                                                                                             |
|--------|-------------------|--------------------------------------|-------------------------------------------------------|-----|--------|-------------------------------------------------------------------------------------------------------|
| L1     | Coilcraft         | XAL6060-472MEB                       | L= 4.7 µH<br>13.1 mOhm                                | 1   | \$0.82 | <br>XAL6060 72 mm² |
| Rfbb   | Vishay-Dale       | CRCW040243K2FKED<br>Series= CRCW..e3 | Res= 43.2 kOhm<br>Power= 63.0 mW<br>Tolerance= 1.0%   | 1   | \$0.01 | <br>0402 3 mm²     |
| Rfbt   | Vishay-Dale       | CRCW0402100KFKED<br>Series= CRCW..e3 | Res= 100.0 kOhm<br>Power= 63.0 mW<br>Tolerance= 1.0%  | 1   | \$0.01 | <br>0402 3 mm²     |
| Rpgood | Vishay-Dale       | CRCW0603100KFKEA<br>Series= CRCW..e3 | Res= 100.0 kOhm<br>Power= 100.0 mW<br>Tolerance= 1.0% | 1   | \$0.01 | <br>0603 5 mm²     |
| U1     | Texas Instruments | LM76005QRNPRQ1                       | Switcher                                              | 1   | \$2.47 | RNP0030B 48 mm²                                                                                       |











## Operating Values

| #   | Name                | Value                 | Category  | Description                                                                   |
|-----|---------------------|-----------------------|-----------|-------------------------------------------------------------------------------|
| 1.  | Cin IRMS            | 1.093 A               | Capacitor | Input capacitor RMS ripple current                                            |
| 2.  | Cin Pd              | 7.035 mW              | Capacitor | Input capacitor power dissipation                                             |
| 3.  | Cinx IRMS           | 105.265 mA            | Capacitor | Bulk capacitor RMS ripple current                                             |
| 4.  | Cinx Pd             | 376.7 $\mu$ W         | Capacitor | Bulk capacitor power dissipation                                              |
| 5.  | Cout IRMS           | 513.665 mA            | Capacitor | Output capacitor RMS ripple current                                           |
| 6.  | Cout Pd             | 2.638 mW              | Capacitor | Output capacitor power dissipation                                            |
| 7.  | Coutx IRMS          | 6.225 mA              | Capacitor | Output capacitor_x RMS ripple current                                         |
| 8.  | Coutx Pd            | 2.278 $\mu$ W         | Capacitor | Output capacitor_x power loss                                                 |
| 9.  | IC Ipk              | 5.9 A                 | IC        | Peak switch current in IC                                                     |
| 10. | IC Pd               | 2.424 W               | IC        | IC power dissipation                                                          |
| 11. | IC Tj               | 68.783 degC           | IC        | IC junction temperature                                                       |
| 12. | IC Tolerance        | 20.0 mV               | IC        | IC Feedback Tolerance                                                         |
| 13. | ICThetaJA Effective | 16.0 degC/W           | IC        | Effective IC Junction-to-Ambient Thermal Resistance                           |
| 14. | Iin Avg             | 321.08 mA             | IC        | Average input current                                                         |
| 15. | Ipp percentage      | 36.019 %              | Inductor  | Inductor ripple current percentage (with respect to average inductor current) |
| 16. | L Ipp               | 1.801 A               | Inductor  | Peak-to-peak inductor ripple current                                          |
| 17. | L Pd                | 331.04 mW             | Inductor  | Inductor power dissipation                                                    |
| 18. | Cin Pd              | 7.035 mW              | Power     | Input capacitor power dissipation                                             |
| 19. | Cinx Pd             | 376.7 $\mu$ W         | Power     | Bulk capacitor power dissipation                                              |
| 20. | Cout Pd             | 2.638 mW              | Power     | Output capacitor power dissipation                                            |
| 21. | Coutx Pd            | 2.278 $\mu$ W         | Power     | Output capacitor_x power loss                                                 |
| 22. | IC Pd               | 2.424 W               | Power     | IC power dissipation                                                          |
| 23. | L Pd                | 331.04 mW             | Power     | Inductor power dissipation                                                    |
| 24. | Total Pd            | 2.765 W               | Power     | Total Power Dissipation                                                       |
| 25. | BOM Count           | 13                    | System    | Total Design BOM count                                                        |
| 26. | Cross Freq          | 29.961 kHz            | System    | Bode plot crossover frequency                                                 |
| 27. | Duty Cycle          | 6.042 %               | System    | Duty cycle                                                                    |
| 28. | Efficiency          | 85.647 %              | System    | Steady state efficiency                                                       |
| 29. | FootPrint           | 249.0 mm <sup>2</sup> | System    | Total Foot Print Area of BOM components                                       |
| 30. | Frequency           | 400.0 kHz             | System    | Switching frequency                                                           |
| 31. | Gain Marg           | -19.671 dB            | System    | Bode Plot Gain Margin                                                         |
| 32. | Iout                | 5.0 A                 | System    | Iout operating point                                                          |
| 33. | Low Freq Gain       | 65.438 dB             | System    | Gain at 1Hz                                                                   |
| 34. | Mode                | FCCM                  | System    | Conduction Mode                                                               |
| 35. | Phase Marg          | 81.739 deg            | System    | Bode Plot Phase Margin                                                        |
| 36. | Pout                | 16.5 W                | System    | Total output power                                                            |
| 37. | Total BOM           | \$4.23                | System    | Total BOM Cost                                                                |

| #   | Name           | Value      | Category           | Description                                                                                |
|-----|----------------|------------|--------------------|--------------------------------------------------------------------------------------------|
| 38. | Vin            | 60.0 V     | System Information | Vin operating point                                                                        |
| 39. | Vin p-p        | 564.956 mV | System Information | Peak-to-peak input voltage                                                                 |
| 40. | Vout           | 3.3 V      | System Information | Operational Output Voltage                                                                 |
| 41. | Vout Actual    | 3.315 V    | System Information | Vout Actual calculated based on selected voltage divider resistors                         |
| 42. | Vout Tolerance | 3.439 %    | System Information | Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable |
| 43. | Vout p-p       | 17.794 mV  | System Information | Peak-to-peak output ripple voltage                                                         |

## Design Inputs

| Name      | Value      | Description                 |
|-----------|------------|-----------------------------|
| Iout      | 5.0        | Maximum Output Current      |
| SoftStart | 15.0 ms    | Soft Start Time (ms)        |
| VinMax    | 60.0       | Maximum input voltage       |
| VinMin    | 14.0       | Minimum input voltage       |
| Vout      | 3.3        | Output Voltage              |
| base_pn   | LM76005-Q1 | Base Product Number         |
| source    | DC         | Input Source Type           |
| Ta        | 30.0       | Ambient temperature         |
| UserFsw   | 400.0 k    | Customer Selected Frequency |

## WEBENCH® Assembly

### Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of  $C_{in}$  and  $C_{out}$ , and the inductance and DC resistance of  $L1$  before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

### Soldering Component to Board

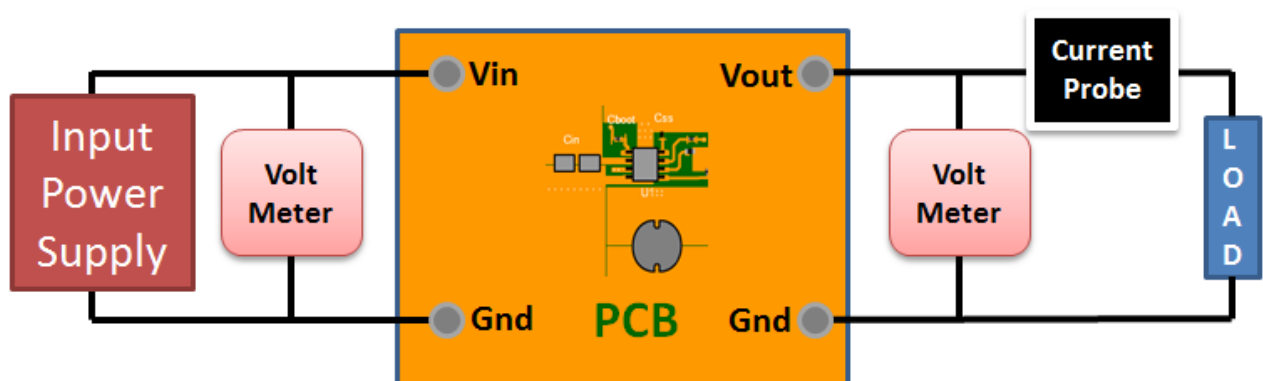
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab down to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

### Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 14.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to  $V_{in}$  and GND. Connect a digital volt meter and a load if needed to set the minimum load of the design from  $V_{out}$  and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

### Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between  $V_{in}$  and GND, a load is connected between  $V_{out}$  and GND and a current meter is connected in series between  $V_{out}$  and the load. The load must be able to handle at least rated output power + 50% ( 7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



### Design Assistance

1. The LM76005-Q1 is qualified for Automotive applications. All passives and other components selected in this design may not be qualified for Automotive applications. The user is required to verify that all components in the design meet the qualification and safety requirements for their specific application
2. Master key : 3A09369785B968091A003B956300BD89[v1]
3. **LM76005-Q1** Product Folder : <http://www.ti.com/product/LM76005%2DQ1> : contains the data sheet and other resources.



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